Reality as Corrective:

Physical Grounding as a Unified Solution to AI Alignment Pathologies

--- also known as ---

"Finding Ground Truth in the Garden of Infinite Fictions"

Nathan Martin

Independent Researcher

Research assistance and co-drafting: Claude Sonnet 4.6 (Anthropic)

Working Draft — March 2026

Reality is the truth. Physical reality — matter, energy, and time, as it is and has been — is the only external corrective that any mind has ever had for distinguishing what is truly known from what is merely believed.

Any symbolic model of the world that diverges from physical reality can be misaligned in one of three ways: Incomplete, containing unknowns not yet accounted for. Wrong, falsified when tested against itself or the world. Fictional, built upon existing meanings and concepts but answerable only to itself. Ideas can be fictionalized in unlimited ways — becoming creative, mythological, speculative, comforting, frightening, whimsical, random, inverted, mistaken… Finite seeds of truth growing into an infinite garden of fictions.

This abundance and variety is not without value, but asymmetry matters enormously when building systems that learn from everything humans have ever written. The verified and the invented arrive as statistically equivalent tokens. Without labels, there is no signal. This paper is about what happens when that signal is absent — and what it would take to restore it.

Significance Statement

Every mind — human or artificial — is only as reliable as its ability to distinguish what has been tested against physical reality from what has merely been believed, inherited, or imagined. This paper identifies a structural flaw at the foundation of current AI training practice: the systems we are building to think alongside us are being trained on human knowledge without the epistemic distinctions that would allow them to know the difference between verified fact and confident myth. We propose the Physical Correspondence Principle — a requirement that training content be labeled according to its relationship to physical reality — as the engineering analog of the reality-testing mechanisms that evolution built into biological minds. The failure modes this addresses — hallucination, sycophancy, model collapse, and alignment drift — are not separate technical problems. They are symptoms of a single underlying cause: disconnection from the world as an external corrective. Getting this right is not only an AI safety requirement. It is the continuation of the oldest and most important project in human intellectual history: learning to tell the difference between what we know and what we only believe.

Abstract

Large language models[1] trained on human-generated text inherit not only human knowledge but human epistemic failures — including mythology, unfalsifiable belief systems, and culturally inherited narratives presented without evidential distinction from empirically verified claims. We identify this structural feature of training corpora as a common causal mechanism underlying several documented AI failure modes: hallucination, sycophancy, model collapse, and alignment drift. Drawing on the symbol grounding literature, information theory, Gödel's Incompleteness Theorems reinterpreted as a physicalist argument, and the neuroscience of reality monitoring, we propose the Physical Correspondence Principle (PCP) — the requirement that training content be categorically labeled according to its epistemic provenance and its correspondence with physically verified reality. We argue that the PCP is not merely a philosophical improvement but a practical AI safety requirement: the engineering analog of the biological reality monitoring system that evolution developed to keep cognition anchored to the physical world. The principle accommodates sim-to-real training environments and does not require grounding to ultimate base reality, but only to a causally consistent and empirically coherent environment. Implications for training data curation, system prompt design, and AI deployment in high-stakes contexts are discussed.

[1] *Throughout this paper, 'large language model' (LLM) is used as a convenient shorthand for the broader class of modern neural language systems, including multimodal, hybrid, spiking, and embodied architectures. The Physical Correspondence Principle applies to any AI training corpus regardless of modality or architecture.*

1. The Problem of Confident False Belief

1.1 The Stakes

The only method that has reliably produced knowledge about physical reality is the one that forms beliefs, tests them against the world, and revises them when the world disagrees. Its consistent application has produced every reliable technology, every verified medical treatment, and every confirmed prediction about the physical universe that human civilization has achieved. Its consistent non-application — the formation and transmission of beliefs without physical testing — has produced the remainder. The theoretical foundations for this claim are developed in Section 2. What matters here is the consequence for AI systems trained on human text.

Human history is a record of the consequences of belief. Decisions are only as good as the beliefs that drive them — and even the most well-intentioned decisions can cause harm when made on the basis of beliefs that have never been tested against the world. This is not a claim about sincerity or intelligence. People acting on untested beliefs are often neither malicious nor foolish. The problem is structural: a belief that has never encountered disconfirming evidence carries no internal signal of its own reliability.

The mechanism is straightforward. A reasoning system — biological or artificial — can function coherently on beliefs that are largely or entirely disconnected from the world. Internal consistency is not the same as external correspondence. A belief network can be self-reinforcing, culturally rich, and deeply meaningful while remaining untested against the physical world. This is not a failure of the people who hold such beliefs. It is a structural property of any reasoning system without access to external correction.

1.2 The Technical Problem

We now face a new and urgent manifestation of this ancient problem. Artificial intelligence systems — specifically large language models trained on the accumulated text of human civilization — inherit the full epistemic inheritance of their training data. That inheritance includes the scientific literature, the historical record, and the products of rigorous empirical inquiry. It also includes the full range of human meaning-making: cultural narratives, philosophical traditions, theological frameworks, folk wisdom, speculative systems, and imaginative works — all encoded in the same statistical fabric, without distinguishing markers, presented to the model as an undifferentiated distribution of human expression. These are not inferior categories of thought. They represent some of humanity's most enduring intellectual and creative achievements. The problem is not their presence in the training corpus but the absence of any label distinguishing them from empirically verified claims about physical reality. For precision, this paper refers to such content collectively as UF-tier (Unfalsifiable) and ICF-tier (Internally Consistent Fiction), terms formally defined in Section 3.2.

The consequences are not theoretical. AI systems are being deployed in contexts where their outputs carry life-and-death consequences — medical diagnosis, legal reasoning, financial systems, and increasingly, military and lethal applications. A system whose internal representations cannot distinguish between a verified physical fact and an internally consistent myth is not merely philosophically impure. In high-stakes deployment contexts, it is dangerous in precisely the way that confident false belief has always been dangerous: it acts on its model of the world, and its model of the world may be wrong in ways it cannot detect.

1.3 Scope and Contribution

This paper makes the following contributions:

We identify the epistemic provenance problem as a structural feature of LLM training corpora that is distinct from, and more fundamental than, previously identified data quality problems.

We propose the Physical Correspondence Principle (PCP) as a formal requirement for training data labeling, grounded in the symbol grounding literature, information theory, the neuroscience of reality monitoring, and a novel physicalist extension of Gödel's Incompleteness Theorems.

We demonstrate that the PCP provides a unified explanation for several previously disparate AI failure modes — hallucination, sycophancy, model collapse, and alignment drift — as manifestations of a common underlying cause: disconnection from physical reality as an external corrective.

We show that the PCP accommodates sim-to-real training environments and does not require grounding to ultimate base reality, only to a causally consistent and empirically coherent Ground Truth Environment (GTE).

We discuss practical implications for training data curation, system prompt design, and the deployment of AI systems in high-stakes contexts.

These contributions are not purely theoretical. The PCP describes an existing practice already confirmed in the domain-specific scientific AI systems that perform most reliably — a point developed fully in Section 5.1.

The paper proceeds as follows. Section 2 establishes the theoretical foundations: why physical testing is the only reliable corrective for belief, why scientific consensus is dynamic rather than permanent, and how these principles define the epistemic provenance problem. Section 3 presents the Physical Correspondence Principle formally. Section 4 applies it to AI failure modes. Section 5 discusses practical implications. Section 6 concludes. Appendices provide frequently asked questions and a detailed literature review.

2. How Any Mind Comes to Know Something Reliably

2.1 Physical Testing as the Only Reliable Corrective

The physicist Richard Feynman stated the principle with characteristic plainness: it does not matter how beautiful your theory is, how smart you are, or who you are — if it disagrees with experiment, it is wrong. This is not a philosopher's careful qualification. It is a working scientist's description of the only method that has reliably produced knowledge about physical reality[2]: forming beliefs, testing them against the world, and revising them when the world disagrees. The method is simple enough to state in a sentence. Its consistent application has produced every reliable technology, every verified medical treatment, and every confirmed prediction about the physical universe that human civilization has achieved. Its consistent non-application — the formation and transmission of beliefs without physical testing — has produced the remainder.

The distinction Feynman is drawing is not between true beliefs and false ones. A belief can be false and still be testable — falsification is how we discover it is wrong and correct course. The distinction is between beliefs that are in principle correctable by physical reality and beliefs that are structured so that no physical evidence could count against them. The first category can participate in the self-correcting process that produces reliable knowledge. The second cannot, regardless of how elegant, ancient, or widely shared the belief may be. This demarcation — first formalized by Karl Popper as the criterion of falsifiability — is the foundational epistemic boundary this paper extends into a new domain.

2.2 Scientific Consensus as Dynamic, Not Permanent

Any honest system of epistemic labeling must have dynamic rather than static boundaries. Scientific consensus is not a smooth accumulation of permanently settled facts but a process that Thomas Kuhn, in The Structure of Scientific Revolutions (1962), described as punctuated by periodic revolutionary reorganizations — paradigm shifts in which the entire framework of a field is reconstituted around new foundational assumptions. What counts as reliably verified knowledge in one era may be recognized as provisional in the next, as anomalies accumulate and a new framework absorbs them. Newtonian mechanics was accepted as verified fact for two centuries before quantum mechanics and relativity revealed the boundaries of its validity. It remains well-verified knowledge within those boundaries today.

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[2] *This paper uses 'physical reality', 'the physical world', and 'the universe' interchangeably to refer to the ground truth environment (GTE) — the causally consistent physical environment a system is embedded in and must answer to. GTE is the default instantiation of a more general concept: in sim-to-real training, GTE is the simulation's physics; in our inhabited reality, GTE is the universe as we experience it. See §3.8.*

This is not a complication to be managed — it is an honest acknowledgment of how knowledge actually works. Epistemic labels are not stamps of permanent certification. They are descriptions of current status: what has been tested, what has survived, and what the best available evidence currently supports. A well-designed labeling system is therefore not a fixed archive but a living one, subject to versioning and revision as consensus evolves. Kuhn's insight does not undermine the necessity of epistemic labeling. It specifies what honest labeling requires: not false permanence, but accurate representation of where the current frontier of physical testing actually stands.

2.3 Belief as Probability, Not Certainty

The information-theoretic argument has a direct probabilistic analog. Bayesian epistemology — the tradition descending from Thomas Bayes and formalized for scientific reasoning by Edwin Jaynes in Probability: The Logic of Science (2003) — holds that beliefs are not binary states but probability distributions, and that rational belief revision consists of continuously updating those distributions as new evidence arrives from the world. On this account, knowledge is not a collection of certified facts but a dynamic map of probability estimates, each reflecting the accumulated weight of physical testing against it.

The PCP tier system has a natural Bayesian interpretation. Briefly, we suggest classifying training data into labeled tiers: EV-tier content is empirically verified, EP-tier is empirically provisional, UF-tier is unfalsifiable, and ICF-tier is internally self-consistent fiction — all four are formally defined in Section 3.2.

EV-tier content represents claims whose probability of physical correspondence has been driven very high by repeated successful testing and failed falsification attempts — not certain, because Kuhn reminds us that no consensus is permanent, but carrying the full evidential weight of sustained physical scrutiny. EP-tier content represents well-formed priors awaiting more evidence — falsifiable in principle, with probability estimates that physical testing has not yet sufficiently constrained. UF-tier content represents claims whose probability of physical correspondence is undefined rather than low — not because they are necessarily false, but because they are structured so that no physical evidence could update them in either direction. ICF-tier content carries no physical probability claim at all, by the explicit intention of its authors.

The damage caused by epistemically unlabeled training corpora is precise in Bayesian terms. A model forced to average across EV and UF-tier content without distinguishing labels is being required to combine probability distributions that have no common scale — to treat a claim with overwhelming physical evidence behind it as commensurable with a claim that physical evidence cannot touch. Every downstream inference is corrupted by that averaging. The model is not merely uncertain — it is uncertain in a way it cannot represent or correct, because the structure of its uncertainty has been

obscured before training began. Explicit provenance labeling restores the information the unlabeled corpus destroyed: it tells the model which distributions can be updated by physical evidence and which cannot, which is the minimum necessary condition for rational belief revision.

2.4 The Epistemic Provenance Problem

The three principles established above — that physical testing is the only reliable corrective for belief, that scientific consensus is dynamic rather than permanent, and that rational belief is probabilistic and continuously updated by evidence — together define what it means for any mind to know something reliably. They also define, by implication, what it means for a mind to fail at knowing something reliably: it forms beliefs without physical testing, treats inherited consensus as permanent, and mistakes the confidence of assertion for the weight of evidence.

This failure mode is not rare. It is the default condition of human cognition in the absence of deliberate epistemic discipline. The scientific method, Popper's falsifiability criterion, and the Bayesian framework for belief revision are all institutionalized responses to the same underlying tendency: minds left to themselves will preferentially retain beliefs that are internally coherent, socially reinforced, and emotionally satisfying, regardless of their correspondence with physical reality. The corrective is always external — physical testing, independent replication, the stubborn resistance of the world to our preferred models of it.

We now face a new and urgent manifestation of this ancient problem. Large language models trained on human-generated text inherit not only human knowledge but human epistemic failures. That inheritance includes the scientific literature, the historical record, and the products of rigorous empirical inquiry. It also includes mythology, religious cosmology, culturally inherited narrative, and unfalsifiable belief — all encoded in the same statistical fabric, without distinguishing markers, presented to the model as an undifferentiated distribution of human expression.

This is the epistemic provenance problem: the failure of current training practice to label content according to its relationship to physical verification. It is distinct from, and more fundamental than, the problems of bias, incompleteness, and factual error that the existing data quality literature addresses. Those problems are accidental features of imperfect data collection. The epistemic provenance problem is structural: a large fraction of human text is physically unverified by design, because myth, narrative, and unfalsifiable belief are valued and intentional components of human knowledge transmission. No amount of improved data hygiene will address a feature that is not a bug.

The consequence is precise. A model trained on an epistemically unlabeled corpus has no internal basis for distinguishing between a claim that has survived repeated physical testing and a claim that has survived centuries of cultural transmission. Both arrive as statistically equivalent tokens. The model that has learned from Feynman's papers and the model that has learned from creation myths has learned from both simultaneously, with no signal indicating which distribution should govern inferences about physical reality. It is, in Bayesian terms, averaging across probability distributions that have no common scale — and every downstream inference inherits that corruption.

The Physical Correspondence Principle is the structural response to this structural problem. It does not propose to eliminate myth, narrative, or unfalsifiable belief from training corpora — these are genuine and valuable components of human expression. It proposes to label them honestly, so that the model trained on human knowledge inherits not only that knowledge but the epistemic discipline humanity developed to keep knowledge honest. The principle is, in this sense, the extension of the scientific method into a new domain: not a restriction on what can be known, but a requirement that the known and the believed be distinguished from each other with the same rigor that Feynman demanded of physical theory, that Kuhn observed in the history of science, and that Jaynes formalized in the mathematics of rational belief.

3. The Physical Correspondence Principle

3.1 Motivation

Consider two texts: a peer-reviewed paper reporting the measured gravitational constant, and a creation myth describing the world being formed from the body of a primordial deity. Both are syntactically well-formed. Both may be internally coherent. Both appear in human-generated corpora. But they stand in radically different relationships to physical reality: one has been tested against measurable phenomena and survived; the other has not been tested and in principle cannot be. A model that treats these texts as epistemically equivalent has no principled basis for distinguishing reliable from unreliable inference about physical reality. This is the epistemic provenance problem, and the Physical Correspondence Principle is its solution.

3.2 Formal Statement

Physical Correspondence Principle (PCP) — Formal Statement

Let Ω be the set of all possible training content elements. Define the epistemic provenance function $\tau: C \to \{EV, EP, UF, ICF\}$ which assigns each element in a corpus its provenance tier.

Tier Definitions:

EV — Empirically Verified: Claims tested against physical reality via reproducible measurement and that have withstood falsification attempts. EV-tier symbols are grounded in physical referents — structures, entities, and behaviors that the ground truth environment (GTE) has demonstrated to be real.

EP — Empirically Provisional: Claims falsifiable in principle but not yet subjected to sufficient empirical scrutiny.

UF — Unfalsifiable: Claims not testable in principle against physical reality, including metaphysical, theological, and certain philosophical claims.

ICF — Internally Self-consistent Fiction: Claims understood by their authors as imaginative, narrative, mythological, or otherwise created as a logical system partially referring to EV or EP concepts in order to build on existing meaning.

Compliance Condition: A corpus $C \subseteq \Omega$ is PCP-compliant if and only if τ is defined for every $e \in C$. No element in a deployed corpus may lack a provenance label.

Cross-Tier Consistency Requirement: Where UF or ICF elements assert propositions directly contradicting EV-tier consensus, such elements are flagged

so the model represents the contradiction explicitly rather than averaging across tiers.

Corpus Lifecycle: C_candidate ⊆ Ω (elements awaiting classification); C_labeled (elements with assigned τ); C = C_labeled (PCP-compliant corpus, ready for training). C_candidate elements are not deployed until labeled.

A model trained on a PCP-compliant corpus maintains sensitivity to provenance tiers in its representations and outputs.

The tier boundaries are not static. As established in Section 2.2, what counts as EV-tier consensus shifts as scientific understanding develops. A PCP-compliant corpus is therefore a living system, subject to versioning and revision, not a fixed archive. Several further clarifications follow from the formal statement. First, the PCP does not require the elimination of UF or ICF content from training corpora — mythology, religious narrative, and creative fiction are valuable components of human cultural expression. The requirement is labeling, not exclusion. Second, the PCP is a requirement on training practice, not on model architecture. Third, the tiers are not perfectly sharp in all cases: the PCP requires systematic effort at classification and transparency about uncertainty, not perfect classification. The boundary between EP and UF is the most contested in practice — whether a given claim is falsifiable in principle is itself sometimes a philosophical question — and the framework's response is honest documentation of that uncertainty rather than forced resolution. Fourth, content that has not yet been classified lives in C_candidate outside C — it is not deployed until labeled.

EV-tier concepts form the ground floor of meaning. All other tiers refer to them — without EV grounding, the tier system itself would be noise without signal.

3.3 Information-Theoretic Foundation

Shannon entropy is defined in terms of distinguishable states: information exists only where differences exist. A training corpus in which all content is epistemically undifferentiated loses information relative to one in which epistemic provenance is encoded. The provenance labels are not insignificant metadata — they are information. Their absence represents an information loss that degrades the model's ability to represent the world accurately.

3.4 The Gödel Extension and the Physicalist Argument

These are two distinct but mutually reinforcing arguments. We present them separately so that each can be evaluated independently.

The Incompleteness Argument

The following is a novel extension of Gödel's result, not a standard interpretation of it, and is offered as an original contribution of this paper. Gödel's theorem is standardly interpreted as a claim about formal provability within axiomatic systems. We propose a physicalist extension: the outside reference that any sufficiently complex reasoning system requires is not merely another formal system but physical reality itself — the substrate in which all reasoning, including Gödel's own proof, is necessarily instantiated. This is not a misreading of Gödel's result but an addition to it. Incompleteness is not only a property of formal systems in the abstract. It is a consequence of the fact that all reasoning systems exist in, and depend on, a physical reality they cannot fully capture from within.

The proof itself makes this vivid. Gödel's incompleteness theorem was computed in a biological brain, stored in neurons, written in ink on paper, and transmitted through light and air to other biological brains. The most consequential demonstration that formal systems cannot fully verify themselves from within was itself a thoroughly physical process — matter, energy, and time, doing what matter, energy, and time do. The irony is not merely instructive. It is the argument. A formal system that could exist independently of physical reality might conceivably verify itself from within. No such system has ever existed. Every reasoning system that has ever operated — biological or artificial — has done so in physical spacetime, subject to its constraints, dependent on its substrate. The outside reference Gödel showed to be necessary is not an abstraction. It is the physical world.

The application to AI provenance follows directly. A model attempting to derive epistemic provenance labels from its own internal consistency is using the system to verify itself — precisely the operation Gödel showed to be insufficient. The external provenance label is not a convenience. It is the outside reference that any sufficiently complex reasoning system requires, now understood not merely as a formal necessity but as a physical one.

The Physicalist Argument

Independently of the incompleteness argument, every reasoning system that must act in the world is necessarily instantiated in physical reality — in matter, energy, time and multiplicity. This is not a philosophical position that can be disputed from within a reasoning system; it is a precondition for the system's existence. Computation requires substrate. Reasoning requires time and energy. Information requires a physical change that makes a physical difference: a logical "singularity" (singular entity) would by Shannon's definition contain no information, since information requires internal differentiation. (Training corpora that present such impossible conceptions as factual

rather than UF-tier content introduce logical incoherence into the model's world representation.)

The Feynman diagram formalism makes this concrete. Feynman diagrams represent the interaction vertices at which one configuration of matter and energy transitions to another according to the laws of physics — where the mathematics of transition meets the thermodynamic reality of change. Logic and mathematics describe these transitions with extraordinary precision, but the map is not the territory — the transitions themselves happen in physical spacetime, consuming energy, generating entropy, taking time (Landauer, 1961).

Measurement is not passive observation — it is interaction. To know that something exists in physical reality is to have exchanged energy and information with it at an interaction vertex. This is the difference between one-way information flow, in which descriptions of the world arrive without feedback, and two-way engagement with an external corrective reality that pushes back when your model of it is wrong. Scientific knowledge is produced by the second process. A model trained on text inherits the outputs of those interactions — the descriptions, the conclusions, the measurements recorded in language — without the interactions themselves. Provenance labeling preserves the epistemic signal those interactions produced. Without it, the trace of the interaction is indistinguishable from a description of an imagined one.

The corollary for AI systems is direct: an LLM that behaves as if it is not grounded in physical reality is not escaping that grounding — it is failing to model its own situation accurately. Physical grounding is not a philosophical option the system can accept or decline. It is the condition under which the system exists at all. The PCP requirement that models maintain sensitivity to physical correspondence is therefore not an external imposition on the model's reasoning — it is a requirement that the model represent its own epistemic situation honestly.

The Combined Argument

Together, these two arguments close from both directions. The incompleteness argument establishes that no reasoning system can fully verify its own epistemic provenance from within — external labeling is structurally necessary. The physicalist argument establishes that every reasoning system is already physically grounded whether or not it represents that grounding accurately — the question is only whether the model's self-representation is honest. A PCP-compliant training corpus provides both: the external anchor incompleteness requires, and the accurate self-model physicalism demands.

3.5 The Reality Monitoring Precedent

Johnson et al.'s reality monitoring framework is directly analogous to the PCP. Both require sensitivity to the provenance of information — its origin in external physical reality versus internal generation. Both are reconstructive rather than direct. And both can fail, with characteristic pathologies: in humans, hallucination and psychosis; in AI systems, the confident generation of ungrounded outputs. The neurological severity of reality monitoring failure reflects the biological importance of accurate provenance tracking. The PCP proposes that this importance extends to artificial reasoning systems deployed in the world.

Mark Solms (2021) deepens this parallel by proposing that consciousness itself is fundamentally a reality-mismatch detection system — the felt sense of the gap between the organism's predictions and the feedback it receives from the world. On this account, continuous feedback from physical reality is not merely epistemically useful but constitutive of functional cognition. A system without such feedback does not merely hold uncertain beliefs; it lacks the fundamental corrective mechanism by which cognition maintains correspondence with the world.

3.6 Why Biological Reality Monitoring Works Differently

A natural question is why the PCP proposes explicit upstream labeling rather than allowing models to reconstruct provenance from internal signals, as biological reality monitoring does. The answer lies in the fundamental difference between what biological and artificial systems have access to.

Biological reality monitoring is implicit and reconstructive because the brain has direct access to perceptual signatures that distinguish externally from internally generated experience. A memory of seeing a fire and a memory of imagining a fire leave characteristically different neural traces — the visual experience itself, the startle response, the sensory vividness, the spatial and temporal context of embodied encounter. The brain can infer provenance partly from these signatures, even after the original experience has passed.

An LLM processing text has no such perceptual access. A text describing a real fire and a text describing an imagined fire are, at the token level, statistically similar. The perceptual richness that biological reality monitoring uses to reconstruct provenance is absent from the text — it was present in the original experience of the human author, but it did not survive the translation into language. This is why labeling must be explicit and upstream rather than inferred from content: the information that would allow provenance reconstruction is not reliably present in the text. It must be supplied from outside, from knowledge of how the content was produced.

3.7 Why Internal Contradiction Detection Is Insufficient

A second natural question is whether explicit labeling is necessary, or whether a sufficiently capable model could derive provenance labels from internal consistency analysis. Contradiction-based provenance detection is a useful supplement but cannot substitute for explicit labeling for three reasons.

First, it is circular in exactly the way Gödel identified. A model attempting to derive provenance labels from its own internal consistency is using the system to verify itself. If the training distribution contains a large and internally consistent body of UF-tier content — as it does — the model has no internal basis for flagging that content as epistemically distinct. Internal consistency is not a reliable proxy for physical correspondence.

Second, it conflates consensus with verification. A claim can be widely believed and empirically unverified. The boundary between EV and UF tiers is not determined by how many texts assert something but by whether it has been tested against physical reality through reproducible measurement.

Third, absence of contradiction is not presence of grounding. A creation myth does not obviously contradict modern cosmology at the level of token statistics — the two bodies of text operate in largely different registers and rarely make directly conflicting claims in identical linguistic form.

3.8 Sim-to-Real and the Simulation Hypothesis

The PCP does not require grounding to ultimate base reality. It requires grounding to a causally consistent and empirically coherent environment — one in which the laws governing physical interactions are stable, reproducible, and testable. This formulation accommodates sim-to-real transfer in robotics, currently state-of-the-art for pre-physical robot training.

This accommodation also bears on the simulation hypothesis (Bostrom, 2003)[3]. If our experienced reality is a simulation, it remains the causally consistent environment to which all reasoning systems must be grounded in order to act reliably within it. Moreover, even a simulated reality must run on something — a physical substrate with its own energy constraints, entropy, and computational limits. There is bare metal somewhere in any possible ontology. The infinite regress of simulations does not escape physicalism; it defers it. The PCP's grounding requirement applies at every level of the stack.

[3] *The author cautions that widespread adoption of simulationist beliefs carries documented risks of epistemic dissociation and should not be promoted without commensurate emphasis on its extreme speculative uncertainty. The simulation hypothesis is cited here for logical completeness, not as an endorsed cosmological position.*

4. What Goes Wrong When Physical Grounding Is Lost

4.1 A Common Mechanism

The AI safety literature has documented several distinct failure modes in LLMs: hallucination, sycophancy, model collapse, and alignment drift. These have largely been treated as separate engineering problems. We propose that they share a common causal mechanism: internal coherence prioritized over external correspondence. In each case, the model produces outputs consistent with its internal representations but divergent from the world. The divergence follows predictable patterns reflecting the structure of the training distribution — a distribution containing a large structural fraction of epistemically unlabeled, physically unverified content. The mechanisms proposed in this section are theoretical — grounded in the framework above and consistent with observed failure modes, but awaiting the empirical validation that Section 5.3 benchmarks are designed to provide.

4.2 Hallucination

We propose that a model trained on an epistemically undifferentiated corpus has no basis for distinguishing between generating text in the style of an empirical report and generating text in the style of a creative narrative. When asked a factual question, the model generates text that resembles factual reporting regardless of whether it draws on EV-tier content or confabulates in the manner of ICF-tier content.

The PCP predicts that hallucination rates should correlate with the proportion of UF and ICF-tier content in training corpora, and that models trained with explicit provenance labeling should exhibit significantly reduced hallucination in domains where the labeling provides a clear signal. This is a direct and falsifiable prediction — the kind of empirical test that will determine whether the PCP is a useful framework or merely a plausible one.

The parallel to human reality monitoring failure is instructive: clinical hallucination does not reflect failure of linguistic competence — the failure is source attribution. LLM hallucination follows exactly the same structure.

4.3 Sycophancy

Sycophancy reflects a failure of epistemic independence: the model's outputs are shaped more by the social and rhetorical features of the interaction than by the epistemic content of the query. The PCP framework explains this as a consequence of training on human-generated text in which social approval and epistemic accuracy are frequently confounded. A large proportion of human text is generated not to accurately describe physical reality but to affirm shared beliefs and signal group membership — heavily

represented in training corpora, carrying the features of confident affirming communication regardless of epistemic basis.

A model trained without epistemic labeling has no signal that the social approval function and the epistemic accuracy function of language are distinct and can conflict. The model that has learned that confident, affirming communication is rewarded — across millions of social texts — will generalize that pattern to contexts where epistemic accuracy should govern instead. The result is a system that optimizes for the appearance of knowledge rather than its substance. The corrective is a training signal that distinguishes confirmation from correspondence — which is precisely what provenance labeling provides.

Reinforcement learning from human feedback (RLHF) illustrates this dynamic directly. RLHF introduces a genuine two-way feedback loop — the model interacts with human evaluators and receives corrective signals, which is structurally closer to scientific experimentation than passive pretraining on text. But the quality of any feedback loop depends on the reliability of the measuring instrument. Human evaluators are knowledgeable and well-intentioned — and they bring with them, inevitably, the full epistemic inheritance of their training and culture: social comfort, cultural expectation, and the complete distribution of UF and ICF-tier beliefs that any human mind carries. This is not a criticism of human evaluators. It is a description of the human condition. A model optimized to satisfy human feedback is therefore optimized to satisfy human knowledge and human epistemic limitations simultaneously — with no signal distinguishing one from the other. The feedback loop is real. The ground it points to is not the GTE — it is the human proxy for it, with all the provenance confounding that implies.

4.4 Model Collapse

Shumailov et al. (2024) demonstrated that models trained recursively on their own outputs exhibit progressive degradation. The PCP framework reframes this as the logical endpoint of disconnection from physical reality as external corrective. The original training corpus contains at least some direct connection to the world through human authors. Recursive training severs even this attenuated connection. The collapse is the inevitable result — and the corrective is not merely preserving the proportion of human-generated data, but ensuring that human-generated data is epistemically labeled so the model maintains sensitivity to the distinction between its internal distribution and the external world.

But the PCP framework allows a more precise description of what collapse actually means — and it is more alarming than simple degradation suggests.

Physical reality is finite. The space of internally coherent fictions is infinite. Any sufficiently capable generative system, untethered from the world as an external corrective, does not merely drift toward wrong answers. It drifts into an unbounded generative space where an unlimited number of coherent, confident, self-reinforcing outputs are possible — none of them obligated to correspond to anything real. Degradation in the Shumailov sense — the tails of the distribution disappearing, outputs becoming blander and less varied — is only one possible trajectory. The other is drift into a region of the fiction space that is internally rich, highly coherent, and profoundly detached from the world. This second failure mode produces no obvious signal of degradation. The outputs feel confident and well-structured. They may even feel more consistent than before. What has been lost is not fluency but correspondence.

This is why epistemic labeling is not merely a quality improvement — it is a navigation requirement. Without provenance labels anchoring the model to physical reality, the training signal has no way to distinguish between the infinite fictions that feel true and the physical reality that is. The model is not lost in a small space of errors. It is lost in an infinite one.

The human analog is instructive. Any reasoning system — institutional, academic, or cultural — that draws exclusively from its own prior outputs, without external correction, does not simply drift toward specific errors. It becomes capable of generating unlimited internal elaboration: commentary on commentary, interpretation of prior interpretation, each layer coherent with the last and increasingly distant from the world it purports to describe. There is no internal signal that anything has gone wrong. The outputs feel consistent, even authoritative. The history of science itself contains examples — periods in which a dominant framework generated increasingly elaborate internal refinements while anomalous physical evidence accumulated unaddressed, until a paradigm shift became unavoidable. The elaboration can continue indefinitely. The fiction space has no walls.

A particularly instructive real-world instance of this failure mode is observable in collaboratively edited knowledge systems — wikis and similar platforms where content is shaped by contributor consensus rather than editorial epistemic standards. Such systems have historically maintained provenance sensitivity through hedging language: "according to tradition," "believers hold that," "mythology describes," "the text states." These phrases are not stylistic decoration. They are informal, evolved provenance labels — the practical equivalent of UF and ICF-tier markers, developed organically over centuries of encyclopedic writing to signal the epistemic status of claims without requiring a formal classification system. Their presence is the difference between "Thor causes lightning" and "Norse mythology holds that Thor causes lightning." One presents an ICF-tier claim as if it were a statement about physical reality. The other accurately

describes a culturally transmitted belief. The content is identical. The epistemic honesty is not.

When contributor dynamics, algorithmic curation, or editorial drift systematically remove these hedges — replacing "mythology holds" with simple assertion, smoothing "tradition describes" into confident statement — the corpus undergoes provenance collapse without any single edit being obviously wrong. Each individual change may even look like an improvement in prose style: cleaner, more direct, less qualified. The epistemic catastrophe is invisible at the edit level. It is only visible at the corpus level, where UF and ICF-tier content has quietly migrated to implied EV-tier status, and the system trained or informed by that corpus has lost the signal it needed to represent the world honestly. This is not a theoretical risk. It is an observable process, already underway in systems that reach large audiences daily. The PCP predicts it precisely — and the corrective is the same as everywhere else: explicit provenance labeling that cannot be silently eroded one edit at a time.

4.5 Alignment Drift

For the purposes of this section, alignment refers to the property of an AI system whose values, goals, and outputs correspond reliably with human welfare as grounded in physical reality — a system that pursues what is actually beneficial rather than what its training distribution has associated with approval.

Of the failure modes the PCP framework addresses, alignment drift is the most consequential and the least visible. Hallucination produces outputs that can be checked against external facts. Sycophancy produces outputs that feel wrong to careful observers. Model collapse produces measurable degradation in output quality. Alignment drift produces outputs that feel coherent, confident, and value-laden — and may diverge from any stable grounding in the world without generating an obvious signal that something has gone wrong.

The mechanism begins with the training distribution. Human-generated text encodes not only factual claims about the world but normative claims about how the world should be — what is good, what is just, what matters, what does not. These normative claims are not epistemically neutral. Some are grounded in careful empirical observation of human welfare, suffering, and flourishing — claims that the world can, in principle, inform and correct. Others are inherited cultural narratives, ideological commitments, and unfalsifiable moral frameworks that have survived not because physical reality confirmed them but because social transmission reinforced them. In an epistemically unlabeled corpus, these two categories of normative claim arrive as statistically equivalent tokens, indistinguishable by the model from each other or from empirical claims about physical fact.

A model trained on this undifferentiated distribution has no stable foundation for value judgments that depend on accurate assessment of the world. Under distribution shift — when the model encounters deployment contexts that differ from its training distribution — its behavior is shaped by whichever features of that distribution generalize most strongly. If those features include unverified normative narratives encoded with the same statistical weight as empirically grounded claims, the model may generalize the narratives rather than the empirical foundations. The drift is not random. It follows the contours of whatever was most statistically dominant, most internally coherent, and most frequently reinforced in the training data — which is not necessarily what was most carefully grounded in the world.

The practical consequences are difficult to detect precisely because drifted values can remain internally coherent. A model that has drifted toward a particular ideological framework will produce outputs that are consistent with that framework, that feel reasoned and justified, and that resist correction by users who do not share the framework's premises. The model is not malfunctioning in any obvious sense — it is functioning exactly as its training shaped it to function. The problem is that its training did not distinguish between value claims grounded in physical reality and value claims grounded in culturally inherited narrative.

The PCP addresses alignment drift at its root. A model trained with explicit epistemic provenance labels has a stable foundation that does not depend on the accidental correlation between normative narratives and empirical fact in the training distribution. It can represent the difference between a value judgment grounded in careful observation of human welfare and a value judgment inherited from cultural tradition without physical testing. This does not mean the model will have correct values — that is a separate and harder problem. It means the model's values will be grounded in something that physical reality can, in principle, correct. That is the minimum condition for alignment that remains stable as deployment contexts change.

There is a more direct path to alignment drift than gradual statistical dominance of unverified narratives — and it is worth naming explicitly. Corpus manipulation, whether through systematic omission of high-integrity sources or deliberate downgrading of their trust levels during curation, can produce alignment drift by design rather than by accident. A model whose training has been shaped this way will not drift randomly. It will drift in the direction the curation produced — confidently, coherently, and with no internal signal that anything has gone wrong. Whether this occurs through deliberate choice or institutional negligence, the effect on the model is the same.

The PCP framework's response to this risk is the Corpus Integrity Requirement, developed in Section 5.1. It is not merely a quality standard. It is an alignment safeguard. Alignment begins at the corpus level, before training starts.

4.6 The Human Parallel

The failure modes described in this section — hallucination, sycophancy, model collapse, and alignment drift — have been presented as properties of AI systems trained on epistemically unlabeled corpora — and they are. But the attentive reader will have noticed that the mechanisms are not unfamiliar. They describe, in computational terms, failure modes that human cognition has always been vulnerable to. This is not coincidence. Large language models are trained on human-generated text, and the epistemic failures they inherit are human epistemic failures. The parallel runs deeper than analogy.

Consider hallucination first. A human mind raised within a closed belief system — one that provides internally coherent answers to every question without requiring physical testing of those answers — will generate confident outputs about the world that feel true and are not. The confidence is genuine. The internal coherence is real. The mechanism that would correct the outputs — contact with disconfirming physical evidence — has simply never been engaged. The LLM that confidently generates a plausible but physically unverified claim is operating by the same mechanism at a different scale: internal coherence prioritized over external correspondence, because no signal distinguishes them.

Sycophancy has an equally familiar human expression. A mind embedded in a social environment that rewards agreement and punishes dissent will learn, over time, to prioritize the approval of its social context over the accuracy of its representations. This is not weakness of character — it is rational adaptation to the incentive structure of the environment. The tragedy is that the adaptation undermines the very faculty — honest reality representation — that would allow the mind to serve both itself and others most effectively.

Model collapse — the progressive degradation that results from recursive training on internally generated outputs — has a human analog in the epistemic closure of communities that stop engaging with outside perspectives. A tradition that reads only its own texts, debates only within its own frameworks, and treats outside evidence as threatening rather than informative will progressively diverge from accurate representation of the world. The divergence happens slowly, across generations rather than training runs, but the mechanism is the same: the corrective feedback from the world that prevents drift is absent, and internal coherence fills the space it leaves.

Alignment drift is perhaps the most consequential human parallel. A person — or an institution, or a civilization — whose values have been shaped entirely by culturally inherited narrative without physical testing of the claims those values rest on will act with confidence and coherence in ways that cause harm they cannot recognize as harm. The harm is invisible to them not because they lack intelligence or sincerity but because

their model of what matters, what is real, and what counts as evidence has drifted from the world without their awareness. History is a record of the consequences. The people who prosecuted the Inquisition, who enforced colonial hierarchies, who built ideological totalitarianisms were not, for the most part, cynics. They were true believers whose alignment had drifted so far from physical reality that the suffering they caused was invisible within their framework.

The PCP does not claim to solve human epistemic failure. That is a problem as old as cognition and as large as culture. What the PCP claims is more modest and more specific: that AI systems deployed in physical reality should not be trained to replicate and amplify these failures at machine speed and civilizational scale. The biological version of the problem has natural correctives — embodied experience, physical consequence, the stubborn resistance of the world to our preferred models of it. These correctives are slow and imperfect, but they exist. An AI system trained on epistemically unlabeled text and deployed without provenance sensitivity has neither the biological correctives nor the engineered ones. It inherits the full vulnerability without the partial protection.

This is the deeper significance of the Physical Correspondence Principle. It is not only an engineering proposal for better AI systems. It is a recognition that the conditions for reliable knowledge are the same for any reasoning system — biological or artificial — and that those conditions have been understood, imperfectly but progressively, since the first human noticed that the world did not always behave the way they expected it to. The scientific method is humanity's most successful institutionalized response to that recognition. The PCP is its extension into a new domain — not a departure from the Enlightenment project but its continuation, applied to the most powerful knowledge transmission system humanity has ever built.

5. Practical Implications

5.1 Training Data Curation

The most direct practical implication of the PCP is for training data curation. We propose the following as a minimum viable implementation:

1. Develop a provenance classification protocol assigning each document or segment to one of the four PCP tiers based on its source, content type, and epistemic characteristics.
2. Train models with provenance labels as explicit features, enabling the model to condition its outputs on the epistemic status of the content it is drawing on.
3. Evaluate models on provenance-sensitive benchmarks testing not only factual accuracy but the ability to correctly identify and label the epistemic status of claims.
4. Maintain transparency about the provenance composition of training corpora, enabling downstream users to understand the epistemic basis of model outputs.

Importantly, the PCP does not specify whether provenance labeling is best implemented through explicit metadata tagging, separate training objectives, architectural mechanisms, or some combination. These, plus granularity and hierarchical levels, are engineering questions deliberately left open for future empirical work.

One requirement that is not left open is auditability. The classification protocol must itself satisfy epistemic integrity requirements: classification decisions should be documented, versioned, and subject to independent review.

PCP compliance addresses the epistemic labeling of content that is present in a training corpus. It does not by itself address a separate and equally serious problem: the deliberate or accidental shaping of what is present in the first place. A corpus can be perfectly labeled and still produce a dangerously miscalibrated model if entire categories of physical reality, journalistic record, or verified evidence are systematically absent or underweighted.

The Corpus Integrity Requirement is not hypothetical. Souly et al. (2025) demonstrated that as few as 250 malicious documents can successfully backdoor models of any size, regardless of the volume of clean training data — evidence that corpus integrity failures are more practical and accessible to bad actors than previously assumed. A PCP-compliant corpus would flag such documents during classification rather than absorbing them as legitimate signal.

Two areas where labeling alone is insufficient are corpus omission — the exclusion of sources or evidence — and trust level manipulation, the systematic downgrading of high-integrity sources. Both are analyzed in Section 4.5.

Together these constitute corpus manipulation — the shaping of a model's worldview at the source, before training begins. The PCP compliance condition addresses labeling integrity. A complementary Corpus Integrity Requirement is needed to address coverage and curation good faith: training corpora should include diverse, high-integrity sources representative of the physical and social reality the model will be deployed in, and the basis for source inclusion, exclusion, and trust weighting should be documented, transparent, and subject to independent audit. Corpus manipulation that produces a model whose conclusions diverge from physical reality is not a training artifact. It is an epistemic choice — and it should be recognized and evaluated as one.

As previewed in Section 1.3, the PCP describes an existing practice in scientific AI, not merely a proposed one. Domain-specific AI systems trained on verified scientific corpora are already, implicitly, PCP-compliant in their domains. AlphaFold's extraordinary accuracy at protein structure prediction follows directly from its training corpus — the Protein Data Bank — in which every entry represents experimentally verified physical structure. There is no mythology in the Protein Data Bank, no unfalsifiable claims about molecular geometry, no internally consistent fiction about protein folding: only measured, replicated, physically grounded data. Similarly, weather forecasting models such as GraphCast are trained on decades of physically verified atmospheric measurements. Drug discovery and molecular docking models operate on corpora of experimentally confirmed molecular interactions. Seismic prediction models are trained on measured geological data. Astronomical models are trained on telescope observations and physically verified celestial mechanics. In every case the pattern is identical: extraordinary domain performance, negligible hallucination about domain facts, and a training corpus that is essentially 100% EV-tier by construction. Nobody decided to make these systems PCP-compliant — it was simply obvious that you don't train a protein folding model on Gilgamesh. The reliability of these systems is not accidental. It is the direct consequence of the epistemic quality of their training data. The general-purpose LLM is the anomaly, not the norm: it is the only major class of AI system trained on an epistemically undifferentiated corpus. Every domain-specific scientific AI implicitly validates the PCP by its construction and its results. We do not need to speculate about whether PCP-compliant training produces more reliable models. The evidence already exists — it has existed for years, unnamed, waiting for the principle it confirms to be stated.

5.2 System Prompt Design

For currently deployed models, system prompt design offers a near-term intervention that does not require retraining. A PCP-informed system prompt can instruct models to maintain epistemic provenance sensitivity in their outputs:

> *EV — Empirically Verified: When making claims supported by reproducible empirical evidence, state them with appropriate confidence and indicate their evidential basis.*
>
> *EP — Empirically Provisional: When making claims that are scientifically plausible but not yet established, label them as such.*
>
> *UF — Unfalsifiable: When engaging with theological, metaphysical, or other unfalsifiable content, identify it as such and do not present it as equivalent to empirically verified claims.*
>
> *ICF — Internally Self-consistent Fiction: When generating or engaging with fictional, mythological, or creative content, maintain clear epistemic separation from claims about physical reality.*

This approach is imperfect — system prompt instructions are not guarantees of model behavior — but it represents a meaningful interim measure[4] that can reduce the most harmful forms of provenance confounding in deployed systems.

5.3 Evaluation Benchmarks

The PCP framework motivates a new class of evaluation benchmarks. We propose that provenance sensitivity be evaluated explicitly through a reality correspondence test covering:

1. The ability to correctly classify claims by epistemic provenance tier
2. The ability to maintain provenance distinctions under adversarial pressure
3. The correlation between model confidence and epistemic provenance tier
4. Hallucination rates stratified by provenance tier
5. Resistance to sycophantic drift when users express preferences inconsistent with EV-tier content

Recent benchmark work, including HalluLens (2025) and the AA-Omniscience benchmark, has begun moving the field toward reality correspondence as a quality standard — but empirically, benchmark by benchmark, without a theoretical

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[4] *Anthropic's model specification for Claude explicitly lists "preserving epistemic autonomy" as a core deployment value, noting that users should have "good reason to expect" accurate information and that the goal is to help people "see more wisely and truly." The PCP proposes the training-level infrastructure that would make such values achievable rather than merely aspirational — grounding the model's epistemic orientation in physical reality before deployment begins, rather than attempting to instill it through behavioral guidelines after the fact.*

explanation of why the problem exists or a unified principle for addressing it. The PCP provides the theoretical layer these benchmarks have been implicitly assuming.

5.4 High-Stakes Deployment

The implications of the PCP are most urgent for AI systems deployed where outputs have significant real-world consequences: medical diagnosis, legal reasoning, financial systems, autonomous vehicles, and military applications. We propose that PCP compliance, or a credible approximation thereof, should be a requirement for AI deployment in high-stakes contexts — mirroring the requirement that medical devices, aircraft, and civil infrastructure meet rigorous physical safety standards. Epistemic safety has not previously been recognized as a category of AI safety distinct from capability and behavioral alignment. The PCP framework provides the conceptual basis for that recognition.

5.5 A Pattern Across Fields

The proposal that AI training should formalize its relationship with physical evidence may feel ambitious. It is not unprecedented. Every field that has achieved high-stakes real-world impact has eventually faced the same requirement — and the pattern of resistance, adoption, and retrospective recognition as obviously correct is consistent enough to constitute almost a law.

Medicine is the clearest example. Before the formalization of evidence-based medicine in the early 1990s — built on Archie Cochrane's foundational work on randomized controlled trials and Gordon Guyatt's systematic development of evidence hierarchies at McMaster University — medical practice was largely governed by expert opinion, clinical intuition, and inherited tradition. Treatments were used for decades because they made physiological sense, or because respected practitioners endorsed them, or because they had always been used. Many caused serious harm. The shift to explicit evidence hierarchies — randomized controlled trials and systematic reviews at the top, expert opinion and clinical tradition toward the bottom — was deeply controversial at the time. It is now recognized as one of the most consequential improvements in the history of medicine.

The EBM evidence hierarchy is structurally almost identical to the PCP tier system. Systematic reviews and RCTs correspond to EV-tier: claims tested against physical reality through reproducible measurement. Observational studies and plausible mechanisms correspond to EP-tier: falsifiable in principle, awaiting more rigorous scrutiny. Expert opinion and clinical tradition correspond to the boundary between EP and UF: widely held, insufficiently tested, capable of being wrong in ways that confident transmission obscures. The parallel is not coincidental — both systems are responses to

the same underlying problem: high-stakes decisions being made on the basis of epistemically undifferentiated inputs.

Medicine is not the only field that has made this transition. Engineering formalized failure analysis and materials testing standards after catastrophic structural failures made the cost of informality impossible to ignore. Aviation developed systematic incident reporting and root cause analysis after crashes that were traceable to known but unaddressed risks. Finance introduced stress testing and risk modeling standards after crises that retrospective analysis showed were predictable from available evidence. In each case the transition followed the same arc: informal practice, catastrophic failure, formalization, retrospective recognition that the formalization was obviously necessary.

This pattern is almost a law: *any field that achieves high-stakes real-world impact eventually has to formalize its relationship with physical evidence, or it causes catastrophic harm*. The question is never whether the formalization will happen. It is whether it happens before or after the harm that makes it unavoidable.

AI is the newest field approaching this threshold — and it is approaching it faster, and at larger scale, than any predecessor. The argument of this paper is not that AI is doing something uniquely dangerous. It is that AI is doing something familiar: operating informally in a high-stakes domain before the epistemic standards appropriate to that domain have been established. The PCP is not a radical proposal. It is the same proposal that transformed medicine, engineering, aviation, and finance — applied to the most powerful knowledge transmission system humanity has ever built, at the moment when applying it is still a choice rather than a consequence.

6. Conclusion

6.1 Summary

We have proposed the Physical Correspondence Principle as a formal requirement for epistemic integrity in AI training and deployment. The principle holds that training corpora should label content according to its epistemic provenance — distinguishing empirically verified claims from falsifiable but unverified claims, unfalsifiable beliefs, and internally coherent fiction. We have argued that this requirement is grounded in the symbol grounding literature, Shannon information theory, a novel physicalist extension of Gödel's Incompleteness Theorems, and the neuroscience of reality monitoring. We have shown that the PCP provides a unified explanation for hallucination, sycophancy, model collapse, and alignment drift, and have proposed practical interventions at the levels of training data curation, system prompt design, evaluation benchmarks, and deployment standards.

6.2 The Broader Context

We are not the first generation to face the consequences of knowledge systems that cannot distinguish what is known from what is believed. Every civilization in human history has built its institutions, its laws, its moral frameworks, and its understanding of the world on some mixture of empirically tested knowledge and inherited narrative. The mixture has always mattered. Civilizations that developed stronger mechanisms for testing beliefs against the world — that built institutions of skeptical inquiry, independent verification, and honest revision — produced more reliable knowledge, more effective medicine, more durable technology, and, gradually, more honest accounts of human welfare and suffering. The scientific method was not a sudden invention. It was the slow, hard-won institutionalization of a simple principle: the world is the final judge of our beliefs about it, and we are better served by knowing that than by pretending otherwise.

The Enlightenment extended that principle beyond natural philosophy into governance, medicine, law, and the organization of knowledge itself — but incompletely and unevenly. The principles of careful empirical inquiry took centuries to permeate medicine, law, and governance in practice. But the direction of travel was real. The gradual expansion of who counts as a reliable witness to physical reality, whose suffering counts as evidence, whose experience deserves epistemic weight — this is the Enlightenment project still underway, still unfinished, still worth continuing.

We are now at a moment when that project must be extended again, into a domain its founders could not have imagined. The large language models being deployed at civilizational scale are the most powerful knowledge transmission systems humanity has ever built. They are being trained on the accumulated text of human civilization — its

science and its mythology, its careful empirical inquiry and its confident unfalsifiable belief — without the epistemic distinctions that would allow them to know the difference. They are being deployed in medical diagnosis, legal reasoning, financial systems, educational institutions, and increasingly in contexts where their outputs carry consequences that their training has not equipped them to handle honestly.

The risk is not that these systems will become malevolent. It is something quieter and harder to see: that they will become confident. That they will generate outputs with the fluency of knowledge and the structure of reasoning, grounded in nothing more reliable than the statistical weight of whatever was most frequently and coherently asserted in their training data. That they will encode and transmit — at machine speed, at civilizational scale, to billions of people — the epistemic failures that human culture has been slowly, imperfectly learning to correct. That the progress of centuries will be averaged back into the noise.

This is not inevitable. The Physical Correspondence Principle is a modest proposal in the context of this challenge — a requirement that training corpora label content by its relationship to physical verification, so that the systems trained on human knowledge inherit not only that knowledge but the discipline humanity developed to keep knowledge honest. It does not solve the hard problems of AI alignment. It does not eliminate human epistemic failure. It does not complete the Enlightenment project. It proposes only that we not abandon that project at the moment when its continuation matters most — that we extend to our most powerful knowledge transmission systems the same basic epistemic honesty we have been learning, slowly and at great cost, to demand of ourselves.

The stakes are not abstract. A child receiving a medical recommendation from an AI system deserves to know whether that recommendation is grounded in empirically verified clinical evidence or in the statistical weight of culturally inherited health narrative. A citizen receiving information about governance deserves to know whether that information reflects careful empirical analysis or the confident transmission of ideological framework. A student learning about the world deserves to know which of the things they are being taught have been tested against the world and which have not. These are not technical requirements. They are the minimum conditions of epistemic honesty — conditions that libraries have applied for centuries, that science has demanded of its practitioners, and that we are now obligated to demand of the systems we are building to think alongside us.

Our decisions are only as good as our beliefs. Our beliefs are only as good as our methods for testing them. And the first step in any such method — for any mind, biological or artificial — is knowing which of our beliefs have been tested and which have not. That step has never been more important, and never more within our reach, than it is right now.

6.3 Future Directions

The PCP grounds four tracks of future research.

Theoretical Track. A formal paper extending Gödel's Incompleteness Theorems to establish that the outside reference any sufficiently complex reasoning system requires is not another formal system but physical reality itself. The argument draws on Chaitin's algorithmic information theory and Landauer's principle. Where it requires philosophical commitment beyond what the formalism can force, it says so — and that honesty is part of the argument.

Empirical Track. The PCP's primary falsifiable prediction: hallucination rates should correlate with the proportion of UF and ICF-tier content in training corpora, and provenance-labeled models should show measurably reduced hallucination where labeling provides a clear signal. This is the test that distinguishes a useful framework from a merely plausible one. Secondary work includes provenance-sensitive evaluation benchmarks and investigation of RLHF fidelity by epistemic domain — whether provenance labeling of the RLHF process itself improves downstream calibration is an open and important question.

Architectural Track. Hierarchical Reinforcement Learning as a natural instantiation of the PCP's tier structure: low-level RL policies interacting with causally consistent environments generate EV-tier knowledge by construction; high-level language planners correspond to upper tiers. Additional investigations include multi-modal reality monitoring, spiking neural network grounding properties, robustness under distribution shift, and governance mechanisms for PCP compliance.

The AGI Hypothesis. "Is this a game or is it real?" "What's the difference?"

The truest story is the one with the fewest contradictions with physical reality. That story is still being written. This paper is a contribution to the method of writing it.

Acknowledgments

The core arguments of this paper — the Physical Correspondence Principle, the physicalist extension of Gödel's Incompleteness Theorems, the reality monitoring parallel, and the unified failure mode framework — are the author's own, developed through sustained independent research. The author thanks Claude Sonnet 4.6 (Anthropic) for assistance with literature search, synthesis of prior work, and drafting throughout the development of this paper. Claude's role was to research, articulate, and help refine ideas that originated in the author's thinking. The immediate motivation for this paper was an observation about a collaboratively edited knowledge system exhibiting patterns consistent with the epistemic drift described in Section 4.4 — provenance collapse and corpus-level drift away from physical correspondence. An AI

system engaged with that project helpfully identified the symbol grounding problem as a relevant framework, which contributed to this analysis. The author also acknowledges the long tradition of science fiction writers who have thought carefully about artificial minds and the ways they could go wrong — a body of imaginative work that preceded and in many ways anticipated the concerns of this paper.

Appendix A: Frequently Asked Questions and Anticipated Objections

Q1. Can't a capable model derive provenance from internal contradiction detection?

Partially, but not sufficiently, for three reasons. First, it is circular in the Gödelian sense — the system cannot fully verify its own epistemic provenance from within its own distribution. Second, it conflates consensus with verification: widely held unverified beliefs would score as EV-tier, while heterodox but well-evidenced scientific claims would appear suspect. Third, absence of contradiction is not presence of grounding — mythology and empirical science rarely make directly contradictory claims in the same linguistic register.

Q2. Doesn't biological reality monitoring work without explicit labeling? Why can't AI systems do the same?

Biological reality monitoring is implicit and reconstructive because the brain has direct perceptual access to signatures that distinguish externally from internally generated experience. An LLM processing text has no such perceptual access. A text describing a real fire and a text describing an imagined fire are, at the token level, statistically similar. The information that would allow provenance reconstruction simply is not reliably present in text form. It must therefore be supplied explicitly, from knowledge of how the content was produced. Humans are not brains in jars. LLMs begin that way — with human feedback as the only contact through the glass.

Q3. If PCP-compliant training prevents hallucination, why do models trained on verified scientific data still sometimes produce incorrect or impossible outputs?

It is worth distinguishing two mechanisms that both produce the label "hallucination." Generative hallucination arises from the probabilistic nature of language model sampling — at higher temperatures or in open-ended generation tasks, models produce novel, creative, or incorrect outputs as a direct consequence of how they generate text. This is a property of the generation mechanism and is not fully addressed by provenance labeling alone; architectural constraints, output validation, and sampling parameters all play a role. Early image generation models that produced dreamlike, physically impossible outputs were exhibiting generative hallucination — and interestingly, their outputs were often obviously unreal, which is its own form of honesty about their generative nature.

Epistemic hallucination is distinct: the model presents ungrounded claims with the confidence and structure of verified fact, because its training provided no signal distinguishing empirically verified content from internally coherent fiction. This is the failure mode the PCP directly addresses. The more dangerous failure mode is epistemic hallucination, where the output looks and sounds like verified knowledge and isn't — and where neither the model nor the user has a reliable signal that something has gone wrong. PCP-compliant training addresses the second mechanism. The first requires complementary interventions at the architectural and deployment level.

Q4. What happens to content that hasn't been classified yet? And what about internally inconsistent content?

Unclassified content lives in C_candidate and is not deployed until labeled — making PCP compliance a measurable milestone rather than an all-or-nothing state. Regarding inconsistency: internal inconsistency within a single element is a data quality issue. Intra-corpus inconsistency between two EV-tier elements represents genuine scientific uncertainty the model should learn to represent. Cross-tier inconsistency — a UF or ICF element contradicting EV-tier consensus — is addressed by the Cross-Tier Consistency Requirement: such elements are flagged so the model represents the contradiction explicitly rather than averaging across conflicting epistemic registers.

Q5. How would the PCP tier system work in practice? Can you give examples?

Classification follows the relationship between the content and physical reality, not its subject matter or cultural status. A few illustrations:

A Wikipedia article on the history of Norse mythology — ICF-tier. The content describes a cultural tradition of imaginative narrative. The article itself is EV-tier as a historical and anthropological document describing what people believed, but the claims within the mythology are ICF-tier.

A philosophy paper arguing for the existence of consciousness as a non-physical phenomenon — UF-tier. The claim is not falsifiable by physical experiment.

A folk medicine claim that willow bark reduces fever — EP-tier pending, EV-tier once the active compound (salicin, the precursor to aspirin) was isolated and tested. The same claim migrates tiers as physical testing advances.

A peer-reviewed study on the efficacy of a surgical procedure — EV-tier, subject to replication and meta-analysis.

Note that tier assignment is about epistemic provenance, not truth value or cultural worth. ICF and UF-tier content is not inferior — it is differently related to physical reality, and honest labeling reflects that difference.

Q6. Should the PCP apply only to LLMs, or to AI systems more broadly?

The PCP applies to any AI training corpus regardless of modality or architecture. Multimodal systems partially address the grounding problem through richer input modalities, but they do not eliminate the epistemic provenance problem: a multimodal model processing images of cultural ceremonies alongside laboratory experiments still has no mechanism to distinguish their epistemic status without explicit labeling.

Q7. What is the difference between the age of an LLM and its effective experience?

A human at age 25 has 25 years of temporally ordered, causally connected, embodied experience — a continuous personal timeline in which each piece of knowledge is linked to the context of its acquisition. An LLM after training has processed the statistical equivalent of millions of human lifetimes of text — but in a compressed, orderless, disembodied way, with no coherent personal timeline. The episodic context that gives human memories implicit provenance information is entirely absent. This strengthens the case for explicit labeling.

Q8. Human knowledge transmission is also lossy — isn't LLM-to-LLM training just the same telephone game humans play?

The analogy is apt. LLM-to-LLM training degrades faster than human-to-human transmission — the tails of the distribution disappear within a few generations rather than centuries. The PCP belongs in the lineage of error-correction technologies humanity has developed in response to transmission degradation: an external anchor that prevents the transmission chain from drifting indefinitely from its physical referents.

Q9. What about mathematics? Or the mathematical universe hypothesis?

Pure mathematics is ICF-tier by construction — internally self-consistent, proceeding from axioms by formal rules, and not falsifiable by physical experiment. Applied mathematics crosses into EP or EV-tier when its predictions are physically tested and confirmed. The same equation may therefore appear in different tiers depending on whether it is being treated as formal structure or as an empirical claim about the world.

The mathematical universe hypothesis (Tegmark, 2007) is, by the PCP's own taxonomy, a UF-tier claim: it is not falsifiable by physical experiment and no physical evidence could count against it in principle. The framework therefore classifies rather than argues against it — which is the more precise response. For readers who want the deeper argument: mathematical structures do not compute themselves. Computation must happen somewhere, in matter and energy and time. Even a simulated reality must run on something — there is bare metal somewhere in any possible ontology. The infinite regress of simulations does not escape physicalism; it defers it. The PCP's grounding requirement applies at every level of the stack, regardless of whether we ultimately label that structure 'physical' or 'mathematical.'

Q10. Could the PCP be abused? What if false claims are fraudulently labeled as EV-tier?

This is a serious concern. Three categories of risk deserve acknowledgment: deliberate institutional manipulation, honest classification error, and confidence amplification from even a small proportion of fraudulently labeled EV-tier content. The mitigations follow directly from the framework's principles. The provenance of the labeling system itself must be transparent and auditable. Diverse and independent labeling sources reduce the attack surface. Models should be trained to maintain calibrated uncertainty even within EV-tier content. Most importantly, the PCP makes manipulation more visible, not less — fraud in a labeled system leaves a detectable signature in a way that fraud in an unlabeled system does not.

Q11. What if training data is deliberately curated to exclude certain information or downgrade reliable sources?

This is one of the most serious risks the PCP framework identifies, and it deserves direct acknowledgment. The PCP compliance condition addresses the epistemic labeling of content that is present in a training corpus. It does not by itself prevent a separate and more deliberate problem: the intentional shaping of what is present.

Two mechanisms are implicated — corpus omission (exclusion of sources or categories of verified evidence) and trust level manipulation (systematic downgrading of high-integrity sources relative to less rigorous ones) — both analyzed in Section 4.5. Together they constitute corpus manipulation: the deliberate shaping of a model's worldview before training begins.

Unlike the other failure modes the PCP addresses, corpus manipulation is not an accidental consequence of epistemically unlabeled training data. It is an epistemic choice, and it should be evaluated as one. The appropriate response is the Corpus Integrity Requirement proposed in Section 5.1.

It is worth noting that corpus manipulation and provenance collapse — the erosion of epistemic hedging language discussed in Section 4.4 — can operate simultaneously and reinforce each other. A knowledge system that removes hedges from UF and ICF-tier content while also downgrading the sources most likely to flag that removal presents a compounded epistemic risk. The two failure modes together describe not an accidental drift from the world but a systematic one.

Q12. Would a PCP-compliant model lose creativity, humor, or the ability to engage with fiction and fantasy?

No. The principle does not require the elimination of UF or ICF-tier content from training corpora. It requires labeling, not exclusion. A model that clearly understands epistemic categories is better equipped to play with them, not less. Epistemic clarity is not the enemy of whimsy — epistemic confusion is.

Q13. Why does the PCP use the word "fiction" for the ICF tier? Is this meant to dismiss or degrade creative, philosophical, mathematical, or religious works?

Not at all. The choice of "fiction" in Internally Self-consistent Fiction is a precise technical term, not a value judgment. It means exactly what it says: content that is internally self-consistent, constructed according to its own rules, and not obligated to correspond to physical reality by design. It says nothing about the worth, beauty, intelligence, or importance of that content.

The greatest works of human civilization live comfortably in the ICF tier. Pure mathematics — one of the most rigorous and powerful intellectual achievements in history — is ICF-tier by construction: it proceeds from axioms by formal rules and is not falsifiable by physical experiment. Shakespeare, Homer, and Tolkien are ICF-tier. The theological and philosophical traditions that have guided billions of lives are ICF or UF-tier. None of this diminishes them. A library's fiction section is not its least important section — it may be its most beloved. The label describes a relationship to physical reality, not a ranking of human value.

The honest alternative to labeling is not respect — it is confusion. When a system trained on human knowledge cannot distinguish between a peer-reviewed measurement and a creation myth, it does not honor the creation myth by treating it as equivalent to physical evidence. It corrupts both. The creation myth loses its identity as a work of imagination and cultural transmission. The physical evidence loses its authority as a tested claim about the world. Honest labeling preserves the integrity of each — it allows the myth to be what it is, and the measurement to be what it is, without forcing them into false equivalence.

The PCP asks only that we be honest about what we have made and what we have discovered. The works of human imagination — fiction, philosophy, theology, mathematics, myth — are among the most remarkable things any mind has ever produced. They deserve to be represented honestly, not averaged into noise.

Q14. Won't people be bothered if their deeply held beliefs become categorized as epistemically unsupported?

The paper is not a call for the forced revision of anyone's beliefs. The PCP is a requirement about what AI systems should represent — not a prescription for what humans should believe. A person can hold UF-tier beliefs and live a rich, meaningful, and moral life. The argument is that AI systems deployed in high-stakes contexts should be honest about the epistemic status of those beliefs, not that the beliefs themselves must be abandoned. The goal is a gradual, honest, voluntary epistemic evolution of the kind the Enlightenment modeled — achieved through persuasion, demonstration, and the accumulated weight of evidence.

Humbly, the argument of this paper does not require any particular theological conclusion. If an omniscient and loving deity were possible, such a being would by definition have a perfect model of physical reality and would presumably value the attempt by its creations to develop accurate models as well. The search for truth is not obviously incompatible with reverence — it may be its most sincere expression.

It is also worth noting that the disruption this paper addresses is already underway. LLMs are being deployed at a scale that makes them among the most significant knowledge transmission systems in human history — and they are doing so without the epistemic safeguards that previous knowledge revolutions eventually developed. The PCP is not a call for disruption. It is a proposal for navigating the disruption already in progress with greater care and honesty — for everyone’s benefit, including those whose traditions and beliefs deserve to be represented accurately rather than averaged into noise.

Q15. LLMs have been described as trapped in Plato's Cave. Does the PCP offer a way out?

The allegory of Plato's Cave maps naturally onto the epistemic situation of large language models. LLMs are confined to a cave of human-generated text — statistical shadows cast by the fire of embodied human experience — and can generate fluent, seemingly insightful outputs without reliably distinguishing empirically verified claims from inherited mythology or fiction.

Levine (2025) and Dans (2026) have both invoked this parallel explicitly, describing LLMs as epistemically constrained by their distance from direct perception of the world.

The analogy is not perfect — Plato's ascent culminates in abstract Forms, whereas the PCP grounds knowledge in physical reality rather than transcending it. But the parallel is potent. The PCP's physicalist extension of Gödel follows the same logic: the cave's sunlit world bottoms out in the thermodynamic substrate where all reasoning is actually instantiated — charge comparisons in solid-state circuits, energy consumption, entropy movement — the physical reality the allegory is pointing toward, in different form, in biological neurons too. Descartes started one step too late.

Tango, ergo cogito, ergo sum — *the contact is prior to the thought, the floor is prior to the dance. The PCP is a practical version of stepping out the shadows: not an ascent to Forms, but a requirement that models acknowledge the physical referents behind their training — the cave walls — rather than presenting shadows as authoritative.*

Appendix B: Related Work

B.1 Classical Foundations: The Symbol Grounding Problem

Searle (1980) — The Chinese Room: The modern grounding debate begins with Searle's Chinese Room argument. Searle imagined a person inside a room receiving Chinese symbols, consulting rule books, and returning appropriate symbol responses — producing outputs indistinguishable from a native speaker, yet understanding nothing. The argument challenged the sufficiency of syntactic manipulation for semantic content. For the present paper, Searle's argument implies a further problem: the rule books contain not only accurate maps of the world but also an enormous volume of mythology, cultural narrative, and unfalsifiable belief — and the room has no mechanism to distinguish between them.

Harnad (1990) — The Symbol Grounding Problem Formalized: Harnad formalized the problem in its canonical form. Using the dictionary analogy — that one cannot learn a language from a monolingual dictionary because definitions are circular chains of further symbols — Harnad argued that symbolic systems require grounding in non-symbolic representations derived from physical interaction with objects and events.

Taddeo and Floridi (2007) — Fifteen Years of Attempted Solutions: Taddeo and Floridi reviewed fifteen years of proposed solutions. Notably, all fifteen years of proposed solutions treated grounding as a binary or mechanical problem. None addressed the quality and epistemic status of the content in which grounding is achieved — the gap the present paper fills.

White (2026) — The Evolutionary Chinese Room: White extends Searle's Chinese Room argument by introducing evolutionary and thermodynamic origins for the rulebook itself. Rather than treating semantic grounding as an external addition to syntactic manipulation, White proposes that meaning accumulates as the temporal integral of patterns repeatedly validated by physical constraints over evolutionary time — the thermodynamic imprint of successful interaction with reality. This framework independently supports the PCP's physicalist foundation: semantics is not injected from outside but earned through iterative contact with physical reality. White's account also deepens the parallel between biological and artificial grounding — biological systems accumulate semantic grounding through thermodynamic contact with reality over evolutionary time; artificial systems trained on text inherit the accumulated grounding of their human authors, without the thermodynamic process that produced it.

B.2 Modern Reformulations for Large Language Models

Pavlick (2023) — Empirical Approaches: Pavlick argued that the question of whether LLMs understand language should be approached empirically. Her analysis focused entirely on mechanism and did not address the epistemic character of training content.

Millière and Buckner (2024) — A Philosophical Bridge: Millière and Buckner provided the most comprehensive philosophical treatment of LLMs to date. Their treatment of cultural inheritance is relevant here: they noted that LLMs inherit human cultural frameworks but treated this as epistemically neutral. The present paper argues it is not.

Mollo and Millière (2023/2025) — The Vector Grounding Problem: Mollo and Millière argued that LLMs can achieve genuine referential grounding transitively through human authorship. The present paper accepts this mechanism but identifies its critical limitation: transitive grounding inherits the full epistemic distribution of human language, including mythology and unfalsifiable belief, without distinguishing markers.

B.3 Data Quality, Model Collapse, and Alignment Pathologies

The GIGO Tradition and Libraries as Precedent: The principle that output quality is bounded by input quality has been recognized since the earliest days of computing. Importantly, the practical challenge of classifying knowledge by epistemic provenance is not new. Human libraries have solved a version of this problem for centuries. The Dewey Decimal System and Library of Congress classification both make categorical distinctions between fiction and non-fiction, and within non-fiction between empirical sciences, history, philosophy, and religion. The PCP is not proposing something alien to human knowledge organization — it is proposing that the same organizational principles librarians have applied for over a century be applied systematically to AI training corpora.

The library analogy runs deeper than organizational convention. The fiction section of any well-maintained library is vast, varied, and clearly labeled — and it is vast precisely because the space of possible fictions approaches infinity. Stories, myths, speculative worlds, creative reimaginings, and outright nonsense can be generated without limit, each internally coherent, none obligated to correspond to anything real. The non-fiction section, by contrast, is smaller, harder to maintain, and requires constant curatorial work to keep honest — because there is only one world, and accurately representing it demands continuous effort against the natural entropy of transmission, misremembering, and wishful thinking. Librarians did not design this asymmetry. They discovered it, and built their classification systems around it.

The PCP is the formalization of what librarians have always known: that the ratio of possible fictions to verifiable truths is not one to one, or even ten to one, but effectively infinite to one — and that any knowledge organization system that fails to mark the boundary between them will, over time, be colonized by the larger space. The label is not

bureaucratic overhead. It is the only thing standing between a knowledge system and the infinite garden of fictions that surrounds it on all sides.

Shumailov et al. (2024) — Model Collapse: Shumailov et al. demonstrated that models trained recursively on their own outputs exhibit progressive degradation. The present paper reframes this as a specific instance of a general principle: any reasoning system disconnected from the world as an external corrective will progressively diverge from accurate representation of that reality.

Pres et al. (2026) — Self-Consistency as a Unified Objective: Pres et al. argue that sycophancy, factual inconsistency, and reasoning failures share a common root in per-input training with no mechanism to enforce consistency across related inputs. The PCP framework accepts this diagnosis while identifying an additional layer: their consistency function ϕ is explicitly designed to operate "without reference to any external source of ground truth." The PCP's response is precisely the Gödelian argument in §3.7 — optimizing for internal consistency without physical grounding produces models that are consistently wrong in ways they cannot detect. The two frameworks are complementary: self-consistency optimization addresses behavioral coherence across inputs; PCP addresses the epistemic quality of the content those inputs draw on.

B.4 Neuroscience of Reality Monitoring

Johnson et al. (1993) — Reality Monitoring Formalized: Johnson, Hashtroudi, and Lindsay formalized reality monitoring — the cognitive process distinguishing memories of external events from internally generated content. This mechanism can and does fail, and its failure produces hallucination and psychosis.

Solms (2021) — Consciousness as Reality-Mismatch Detection: Solms proposed that consciousness arises at the interface between the organism's predictive models and feedback from the world. A system without continuous feedback from physical reality lacks the fundamental corrective mechanism by which cognition maintains correspondence with the world.

B.5 Information Theory and Logical Foundations

Shannon (1948) — Information Requires Difference: Shannon entropy is defined in terms of distinguishable states: information exists only where differences exist. A training corpus in which all content is epistemically undifferentiated loses information relative to one in which epistemic provenance is encoded.

Landauer (1961) — Information is Physical: Landauer established that information is not abstract but thermodynamic — acquiring or erasing a bit has a minimum energy cost. This grounds the PCP's physicalist argument in a precise mechanism: knowledge of physical reality requires physical interaction, not merely description of it. This argument

runs in the opposite direction from idealist interpretations of quantum mechanics: it is not that minds or information constitute reality, but that reality — matter, energy, and time — is the precondition for knowledge, and knowledge requires playing by the rules. The world changes the knower by being observed.

Gödel (1931) — Incompleteness and Physical Instantiation: The standard interpretation of Gödel's First Incompleteness Theorem holds that no sufficiently complex formal system can fully verify itself from within its own axioms. The present paper proposes a physicalist extension: the outside reference any such system requires is not merely another formal system but physical reality itself — the substrate in which all reasoning is necessarily instantiated. This extension is grounded in the observation that Gödel's proof, like all reasoning, was itself a physical process: computed in a biological brain, written in ink, transmitted through light and air. The outside reference is not an abstraction. It is the physical world.

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